Date	31 October 2026
Team ID	XXXXXX
Project Name	Global Fuel Prices and Taxation Analysis (2020–2026)
Maximum Marks	4 Marks

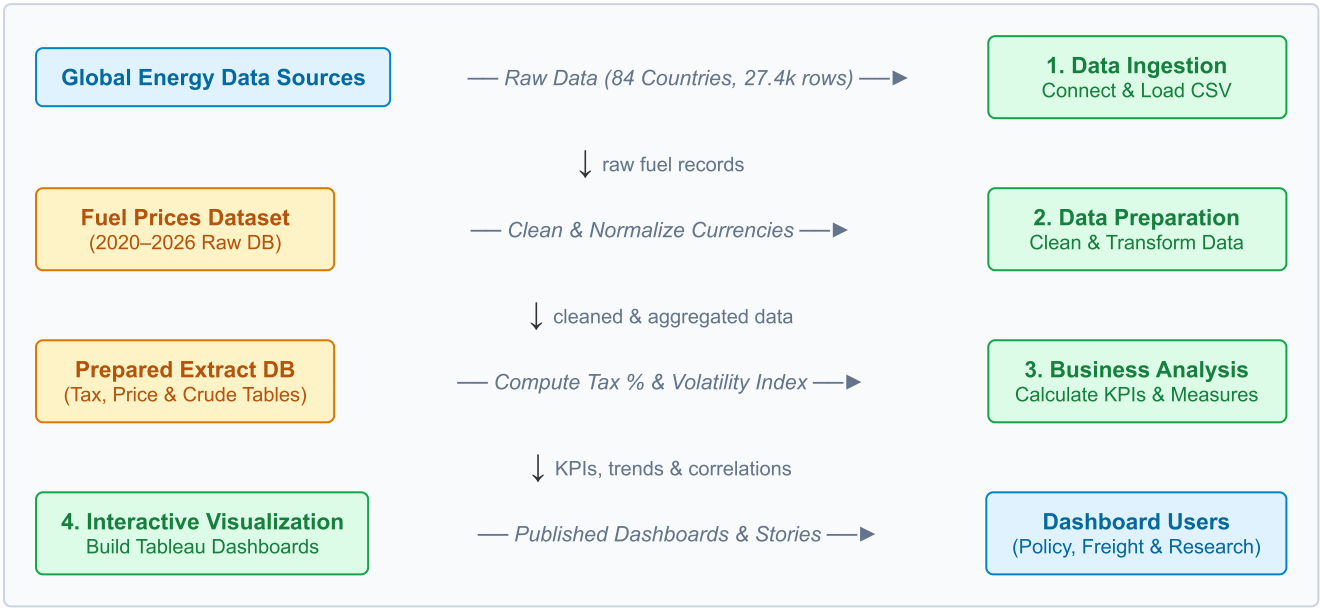
Project Design Phase-II
Data Flow Diagram & User Stories

Data Flow Diagrams:
A Data Flow Diagram (DFD) is a visual representation of the information flows within a system. A clear DFD depicts system requirements graphically by illustrating how data enters and leaves the system, what processes transform the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or data provider outside the system that sends or receives data (e.g., Global Fuel Data Providers, Tableau Users).
Rectangle with header	Process	An activity that transforms incoming data into outgoing metrics (e.g., Data Ingestion, Tax & Subsidy Calculations, Volatility Indexing).
Open/Solid Rectangle	Data Store	A repository where processed fuel data is stored (e.g., Global Fuel Prices Dataset, Cleaned Extract DB).
Labeled Arrow	Data Flow	The movement of data between entities, processes, and stores, labeled with specific data parameters passed.

DFD Level 0: Global Fuel Prices and Taxation Analysis



SMARTBRIDGE
Let's bridge the Gap

SKILL WALLET
A SmartBridge Product

User Stories (Sprint 1 & Sprint 2)

List of functional user stories categorized by persona, epic, priority, and release sprint:

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Pri.	Rel.
Policy Consultant (Dr. Sarah)	Policy & Tax Analysis	USN-1	As a policy consultant, I can view average fuel prices, tax percentages, and subsidy totals as top KPI cards so that I can assess global fiscal patterns at a glance.	KPI cards on Dashboard 1 display accurate global aggregates and update on filter selection.	High	Sp-1
Policy Consultant (Dr. Sarah)	Taxation Analysis	USN-2	As a policy consultant, I can view an interactive tax heatmap across 84 countries so that I can identify high-tax and heavily subsidized fuel markets.	Geographic map visually highlights tax burden levels per country.	High	Sp-1
Policy Consultant (Dr. Sarah)	Structural Price Analysis	USN-3	As a policy consultant, I can evaluate tax percentage versus retail fuel price in a scatter plot so that I can analyze the relationship between taxation and pump prices.	Scatter plot renders tax % on X-axis and retail price on Y-axis with trend lines.	Med	Sp-2
Supply Chain Mgr (Marcus)	Trend Analysis	USN-4	As a supply chain manager, I can track retail prices for Petrol and Diesel over time (2020–2026) so that I can monitor cost trends across logistics networks.	Line charts plot historical multi-fuel price trends cleanly.	High	Sp-1

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Pri.	Rel.
Supply Chain Mgr (Marcus)	Geographic Diesel Mapping	USN-5	As a supply chain manager, I can view geographical distribution maps of diesel prices so that I can identify low-cost refueling regions for freight routes.	Filled map displays color-coded diesel price bands globally.	High	Sp-1

User Stories (Sprint 2 & Sprint 3)

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Pri.	Rel.
Supply Chain Mgr (Marcus)	Forecasting & Volatility	USN-6	As a supply chain manager, I can view predictive fuel price forecasts and regional volatility indices so that I can hedge against budget spikes.	Forecast models display upper/lower confidence intervals.	Med	Sp-2
Energy Economist (Prof. Elena)	Macroeconomics	USN-7	As an energy economics researcher, I can evaluate fuel prices segmented by country income level so that I can study affordability across income brackets.	Bar/box charts group fuel prices by World Bank income tiers.	High	Sp-2
Energy Economist (Prof. Elena)	Subsidy Cross-Tabs	USN-8	As a researcher, I can inspect cross-tabulations of subsidies across regions and income levels so that I can assess fiscal intervention efficiency.	Cross-tabulation table displays subsidy totals and proportions.	Med	Sp-2
Energy Economist (Prof. Elena)	Crude Correlation	USN-9	As a researcher, I can evaluate the correlation between Brent crude spot prices and retail fuel prices so that I can quantify price pass-through dynamics.	Dual-axis/scatter plot calculates correlation coefficients between Brent and retail rates.	Med	Sp-3
All Dashboard Users	Interaction & Filtering	USN-10	As a dashboard user, I can filter by fuel type, region, and year so that all charts update in sync across the entire dashboard view.	Global parameters filter all worksheets cleanly.	High	Sp-2
Dashboard Administrator	Data Pipeline Management	USN-11	As an admin, I can refresh the underlying global fuel dataset seamlessly so that dashboards reflect updated pricing without breaking calculated fields.	Data source refresh completes without missing field errors.	High	Sp-1